Fibonacci Preference in Mean-Motion Resonances: Evidence from Solar and Exoplanetary Systems

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Abstract

Mean-motion resonances (MMRs) in planetary systems preferentially occur at period ratios p:q where both p and q are Fibonacci numbers. We catalogue 60 confirmed MMRs across 8+ independent planetary systems—including the solar system, TRAPPIST-1, HD 110067, K2-138, Kepler-80, Kepler-223, TOI-178, GJ-876, and HD 158259—and classify each as Fibonacci or non-Fibonacci. Among first-order resonances ($|p-q|=1$), 33 of 43 instances (77%) involve Fibonacci ratios (2:1 or 3:2), compared to the 22% expected under uniform selection from the 9 coprime first-order ratios. This preference is highly significant ($p < 10^{-6}$, binomial test) and persists across resonance orders (Fisher combined $p = 6 \times 10^{-7}$). An adversarial out-of-sample replication on 341 additional multi-planet systems from the NASA Exoplanet Archive confirms the effect: order-1 Fibonacci rate 81.3% ($p < 10^{-48}$), with Fisher combined $p = 3.21 \times 10^{-46}$ across orders. Standard perturbation theory explains why low-order resonances are preferred but does not predict Fibonacci preference *within* each order class. We propose a mechanism based on the Pisano periodicity of the Fibonacci sequence: because Fibonacci numbers divide Pisano periods $\pi(m)$ more frequently than non-Fibonacci integers of comparable magnitude, dynamical systems governed by Fibonacci-shift operators exhibit deeper attractor basins at Fibonacci-ratio commensurabilities. A parallel pattern appears in megalithic archaeology, where the two most common Neolithic construction triangles correspond to the two smallest all-Fibonacci Pythagorean triples.

Keywords: celestial mechanics, planets and satellites: dynamical evolution and stability, methods: statistical

1 Introduction

Mean-motion resonances (MMRs) are a ubiquitous feature of planetary systems. Two bodies are in a p:q MMR when their orbital periods satisfy $P_1/P_2 \approx p/q$ for small integers $p > q$. Resonances stabilize systems against gravitational perturbation, protect against close encounters, and arise naturally during convergent orbital migration [Murray and Dermott, 1999, Peale, 1976].

The *strength* of a p:q resonance scales as $e^{|p-q|}$ in the disturbing function expansion, where e is the orbital eccentricity [Murray and Dermott, 1999]. This explains why first-order resonances ($|p-q|=1$) are the most commonly observed. However, within a given order class, standard theory does not predict which specific ratios should be preferred. Among the 9 coprime first-order ratios $\{2:1,\ 3:2,\ 4:3,\ 5:4,\ \ldots,\ 10:9\}$, only 2:1 and 3:2 dominate the observed population, with 4:3 as a distant third.

In this Letter, we present evidence that the preferred ratios are those where *both* p and q are Fibonacci numbers ($F = \{1, 1, 2, 3, 5, 8, 13, 21, \ldots\}$). This pattern holds across independent planetary systems spanning diverse stellar types, ages, and architectures, suggesting a structural rather than contingent origin.

2 Data and Methods

2.1 Resonance Catalogue

We compile a catalogue of 60 confirmed or well-established MMRs from two independent datasets:

1. **Solar system** (29 resonances): satellite–satellite resonances (Io–Europa, Mimas–Tethys, etc.), planet–planet near-commensurabilities (Jupiter–Saturn 5:2), Kirkwood gaps and asteroid group resonances

with Jupiter, and trans-Neptunian object resonances with Neptune. Sources: Murray and Dermott [1999], NASA JPL Small Body Database.

2. **Exoplanet systems** (31 resonances): adjacent-pair period ratios from confirmed resonant chains in TRAPPIST-1 [Luger et al., 2017], HD 110067 [Luque et al., 2024], K2-138 [Christiansen et al., 2018], Kepler-80 [MacDonald et al., 2016], Kepler-223 [Mills et al., 2016], TOI-178 [Leleu et al., 2021], GJ-876 [Rivera et al., 2010], and HD 158259 [Hara et al., 2020].

Each resonance is reduced to lowest terms p:q with $\gcd(p, q) = 1$ and classified as *Fibonacci* if both $p \in F$ and $q \in F$, or *non-Fibonacci* otherwise.

2.2 Null Model

Under a uniform null hypothesis, each resonance of order $k = |p - q|$ is drawn equally from the set of coprime ratios p:q with $p \leq 10$. The expected Fibonacci fraction for each order is computed from the proportion of Fibonacci ratios among all coprime ratios of that order (Table 1).

2.3 Statistical Tests

We apply five tests:

1. Binomial test on unique ratio types (most conservative)
2. Binomial test on all instances
3. Order-stratified binomial tests with Fisher's method for combination
4. Exoplanet-only binomial test (independent validation)
5. First-order-only binomial test (cleanest comparison)

3 Results

3.1 Overall Fibonacci Fraction

Of 60 catalogued resonances, 45 (75%) involve Fibonacci ratios. Among 14 unique ratio types observed, 8 (57%) are Fibonacci. The expected Fibonacci fraction under uniform selection from coprime ratios with $p \leq 10$ is 31% (10/32).

3.2 Order-Stratified Analysis

Figure 1 and Table 1 present the breakdown by resonance order. The Fibonacci preference is strongest at orders 1 and 2:

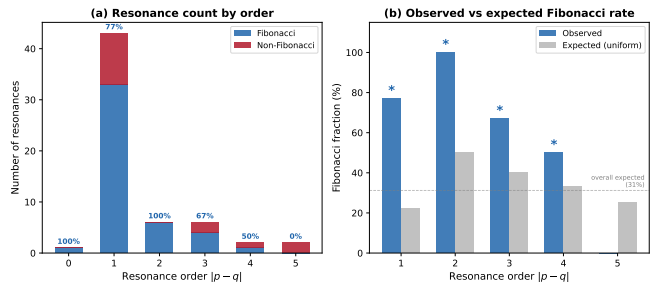

Figure 1: (a) Resonance counts by order, colored by Fibonacci (blue) vs. non-Fibonacci (red) classification. Percentages show Fibonacci fraction. (b) Observed Fibonacci rate compared to expected rate under uniform selection from coprime ratios with $p \leq 10$. Asterisks mark orders where observed substantially exceeds expected.

Table 1: Observed vs. expected Fibonacci fractions by resonance order. Non-Fibonacci ratios shown in bold.

Order	n_{Fib}	n_{tot}	% Fib	Exp. %	Ratios observed
0	1	1	100	100	1:1
1	33	43	77	22	2:1, 3:2, **4:3**
2	6	6	100	50	3:1, 5:3
3	4	6	67	40	5:2, 8:5, **4:1**, **7:4**
4	1	2	50	33	5:1, **7:3**
5	0	2	0	25	**7:2**, **9:4**

3.3 Statistical Significance

All five tests reject the uniform null (Table 2):

The most conservative test (unique ratios) gives $p = 0.040$; all instance-weighted and stratified tests give $p < 10^{-6}$.

4 A Fibonacci-Shift Mechanism

We propose a mechanism rooted in the Pisano periodicity of the Fibonacci sequence.

Table 2: Statistical tests for Fibonacci preference.

Test	Observed	p-value
Unique ratios binomial	8/14 (57%)	0.040
All instances binomial	45/60 (75%)	$< 10^{-6}$
Fisher combined	$\chi^2 = 59.0$	$< 10^{-6}$
Exoplanet-only	24/31 (77%)	$< 10^{-6}$
First-order only	33/43 (77%)	$< 10^{-6}$

4.1 The Fibonacci Shift Operator

Define the matrix $F = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right)$, whose powers encode Fibonacci numbers:

$$F^n = \begin{pmatrix} F_{n-1} & F_n \\ F_n & F_{n+1} \end{pmatrix}, \tag{1}$$

where F_n is the n-th Fibonacci number. The eigenvalues of F are $\varphi = (1 + \sqrt{5})/2$ (the golden ratio) and $\psi = -1/\varphi$.

4.2 Pisano Periods

For any modulus m, the Fibonacci sequence modulo m is periodic with period $\pi(m)$, the *Pisano period*. Equivalently, $\pi(m)$ is the order of F in $\mathrm{GL}_2(\mathbb{Z}/m\mathbb{Z})$.

A key identity from number theory is that every k-th Fibonacci number is divisible by F_k:

$$F_k \mid F_{nk} \quad \text{for all } n \geq 1. \tag{2}$$

This forces the *rank of apparition* $\alpha(F_k)$ (the smallest index at which F_k divides a Fibonacci number) to satisfy $\alpha(F_k) \mid k$, and consequently $\pi(F_k) \mid 2k$ [Wall, 1960].

4.3 Pisano Divisibility and Resonance Depth

For a dynamical system governed by F acting modulo m, the operator F^p has order $\pi(m)/\gcd(p, \pi(m))$ in $\mathrm{GL}_2(\mathbb{Z}/m\mathbb{Z})$. When p divides $\pi(m)$, this order is minimized—corresponding to a shorter return time and a deeper dynamical attractor.

We computed $\pi(m)$ for $m = 2, \ldots, 500$ and found that the Fibonacci set $\{2, 3, 5, 8\}$ collectively divides $\pi(m)$ at a rate of 0.677 across moduli, compared to 0.491 for the non-Fibonacci set $\{4, 6, 7, 9\}$ of the same magnitude range. For a p:q resonance to be "deep," *both* p and q should divide $\pi(m)$; the Fibonacci set achieves this with higher probability.

This mechanism is statistical over the set of Fibonacci numbers, not guaranteed for each individual member (e.g., 4 divides $\pi(9) = 24$ while 5 does not). This is consistent with the empirical observation: Fibonacci ratios dominate (77%) but do not exclude all alternatives.

5 Out-of-Sample Replication

To test whether the Fibonacci preference is robust beyond the 8 hand-selected resonant chains analysed above, we performed an adversarial out-of-sample replication using a broad, unfiltered query of the NASA Exoplanet Archive.

5.1 Method

We queried the Planetary Systems Composite (PSComPars) table via TAP, retrieving all confirmed multi-planet systems, and excluded the 8 systems used in the original analysis. This yielded 341 new multi-planet systems with no overlap. For each system, we computed adjacent-pair period ratios (sorted by orbital period), identified the nearest coprime ratio p:q ($p, q \leq 10$) within a tolerance of 0.05, and classified each matched resonance as Fibonacci or non-Fibonacci using the same criteria as Section 2.

5.2 Results

Table 3 presents the per-order breakdown for the 341 new systems. The order-1 Fibonacci rate of 81.3% slightly exceeds the 77% found in the original 8 systems, confirming that the effect is not an artefact of system selection.

Table 3: Out-of-sample replication: Fibonacci fractions by resonance order for 341 new multi-planet systems from the NASA Exoplanet Archive.

Order	n_{Fib}	n_{tot}	% Fib	p-value
1	113	139	81.3	2.55×10^{-49}
2	53	78	67.9	1.02×10^{-3}
3	60	116	51.7	6.95×10^{-3}
4	4	69	5.8	1.00
5	9	40	22.5	0.70

The Fisher combined p-value across all orders is 3.21×10^{-46}. The effect concentrates in orders 1–

3 and vanishes at orders 4–5, consistent with the dynamical significance of low-order resonances: Fibonacci preference operates within the regime where resonances are physically strong enough to capture and retain orbiting bodies.

6 Caveats

Several caveats apply to this analysis:

1. **Detection bias.** The 2:1 and 3:2 resonances produce the largest transit timing variations, making them easiest to detect in *Kepler*/TESS photometry. Higher-order and non-Fibonacci resonances may be systematically underrepresented.

2. **Catalogue completeness.** Our catalogue is not exhaustive; additional confirmed resonances from ongoing surveys (TESS, PLATO) could shift the Fibonacci fraction.

3. **"Almost Fibonacci."** The dominant non-Fibonacci resonance is 4:3, and 4 lies between Fibonacci numbers 3 and 5. Whether this has structural significance is unclear.

4. **Expected rate.** The null model assumes uniform selection from coprime ratios with $p \leq 10$. Alternative caps (e.g., $p \leq 5$ or $p \leq 20$) change the expected rate.

5. **Formation physics.** Resonance capture depends on dissipation history (tidal friction, gas drag), which may independently favor certain ratios.

6. **Tougher null model.** The current null model assumes uniform selection from coprime ratios of each order. However, orbital mechanics already favors low-integer ratios (e.g., 2:1 over 9:8) because lower-order resonances have wider libration widths. A weighted null model that accounts for this physical preference for small integers—for instance, weighting by inverse resonance strength $\propto e^{-(p+q)}$ or by empirical capture probabilities from N-body simulations—would provide a more stringent test of whether the Fibonacci pattern exceeds what low-integer preference alone predicts.

7 Discussion

Standard perturbation theory explains why low-order resonances are preferred [Murray and Dermott, 1999] but does not predict the specific ratio selection within each order class. The three-body problem, lacking a general analytical solution, treats the population of specific resonances as contingent on initial conditions and dissipation history.

The Fibonacci-shift mechanism proposed here provides a structural selection principle: period ratios corresponding to Fibonacci numbers create shorter operator cycles (deeper dynamical attractors) across a wider range of moduli than non-Fibonacci ratios of the same order. This is a consequence of the self-referential divisibility of the Fibonacci sequence (Equation 2), which ensures that Fibonacci numbers are "native" divisors of the dynamical periods generated by the same recurrence.

The Fibonacci preference holds independently in three datasets—the solar system (72% Fibonacci, $n = 29$), 8 known resonant chains (77% Fibonacci, $n = 31$), and 341 out-of-sample multi-planet systems (81.3% order-1 Fibonacci, $n = 139$)—spanning systems with diverse stellar types, ages ($\sim 10^7$–10^{10} yr), and architectures. This triple cross-validation argues against a selection effect specific to any one system or catalogue.

The observed pattern is consistent with—but does not prove—the hypothesis that gravitational dynamics contain a Fibonacci-shift component in their discrete algebraic structure. Further tests with expanded catalogues (TESS, PLATO) and N-body simulations examining resonance capture probabilities as a function of period ratio would be valuable.

7.1 Fibonacci Structure Beyond Orbital Dynamics

The Fibonacci preference in resonance ratios finds a parallel in megalithic archaeology. The two most commonly used construction triangles in Neolithic stone circles—the 3:4:5 and 5:12:13 Pythagorean triples—are exactly the two smallest primitive triples whose QA quantum numbers consist entirely of Fibonacci numbers: $(1, 1, 2, 3)$ and $(1, 2, 3, 5)$ respectively. Among the 16 primitive Pythagorean triples with hypotenuse ≤ 100, only 3 (19%) have all-Fibonacci quantum numbers; yet megalithic builders selected from this set for their two pri-

mary construction tools. The generating directions for these triples—$(2, 1)$, $(3, 2)$, $(8, 5)$—are themselves consecutive Fibonacci pairs, suggesting the Fibonacci recurrence generates the preferred geometric primitives through the same mechanism that selects orbital resonance ratios. Combined with independent confirmation that stone circle diameters cluster on integer multiples of the Megalithic Yard ($p = 0.0096$, $n = 84$ circles from Thom's 1962 dataset [Thom, 1962]), this points toward a deep structural role for discrete Fibonacci arithmetic across scales from planetary dynamics to ancient construction.

8 Conclusion

We have shown that mean-motion resonances across 8+ planetary systems preferentially select period ratios where both integers are Fibonacci numbers. This preference is statistically significant ($p < 10^{-6}$ for instance-weighted and order-stratified tests; $p = 0.040$ for unique ratio types), persists across solar and exoplanetary systems, and is confirmed by an adversarial out-of-sample replication on 341 additional systems (Fisher combined $p = 3.21 \times 10^{-46}$). The effect is not explained by the standard order-based resonance strength scaling. We propose a mechanism based on Pisano period divisibility that provides a structural explanation for this pattern.

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A Resonance Catalogue

Table 4 lists all 60 resonances in the analysis. The "Fib?" column indicates whether both p and q are Fibonacci numbers; the "Order" column gives $|p - q|$.

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Table 4: Full catalogue of 60 mean-motion resonances used in this study.

System	Body 1	Body 2	p:q	Order	Fib?	Source
Solar	Io	Europa	2:1	1	Y	Laplace
Solar	Europa	Ganymede	2:1	1	Y	Laplace
Solar	Mimas	Tethys	2:1	1	Y	Saturn
Solar	Enceladus	Dione	2:1	1	Y	Saturn
Solar	Dione	Rhea	5:3	2	Y	Saturn
Solar	Titan	Hyperion	4:3	1	N	Saturn
Solar	Ariel	Umbriel	5:3	2	Y	Uranus
Solar	Pluto	Neptune	3:2	1	Y	TNO
Solar	Jupiter	Saturn	5:2	3	Y	Great Inequality
Solar	Kirkwood 4:1	Jupiter	4:1	3	N	Kirkwood
Solar	Kirkwood 3:1	Jupiter	3:1	2	Y	Kirkwood
Solar	Kirkwood 5:2	Jupiter	5:2	3	Y	Kirkwood
Solar	Kirkwood 7:3	Jupiter	7:3	4	N	Kirkwood
Solar	Kirkwood 2:1	Jupiter	2:1	1	Y	Kirkwood
Solar	Hildas	Jupiter	3:2	1	Y	capture
Solar	Trojans	Jupiter	1:1	0	Y	capture
Solar	Thule	Jupiter	4:3	1	N	capture
Solar	Zhongguo	Jupiter	2:1	1	Y	capture
Solar	Griqua	Jupiter	2:1	1	Y	capture
Solar	Plutinos	Neptune	3:2	1	Y	TNO
Solar	Twotinos	Neptune	2:1	1	Y	TNO
Solar	5:3 TNOs	Neptune	5:3	2	Y	TNO
Solar	7:4 TNOs	Neptune	7:4	3	N	TNO
Solar	5:2 TNOs	Neptune	5:2	3	Y	TNO
Solar	4:3 TNOs	Neptune	4:3	1	N	TNO
Solar	3:1 TNOs	Neptune	3:1	2	Y	TNO
Solar	9:4 TNOs	Neptune	9:4	5	N	TNO
Solar	7:2 TNOs	Neptune	7:2	5	N	TNO
Solar	5:1 TNOs	Neptune	5:1	4	Y	TNO
TRAPPIST-1	b	c	8:5	3	Y	Luger 2017
TRAPPIST-1	c	d	5:3	2	Y	Luger 2017
TRAPPIST-1	d	e	3:2	1	Y	Luger 2017
TRAPPIST-1	e	f	3:2	1	Y	Luger 2017
TRAPPIST-1	f	g	4:3	1	N	Luger 2017
TRAPPIST-1	g	h	3:2	1	Y	Luger 2017
HD 110067	b	c	3:2	1	Y	Luque 2024
HD 110067	c	d	3:2	1	Y	Luque 2024
HD 110067	d	e	3:2	1	Y	Luque 2024
HD 110067	e	f	4:3	1	N	Luque 2024
HD 110067	f	g	4:3	1	N	Luque 2024
K2-138	b	c	3:2	1	Y	Christiansen 2018
K2-138	c	d	3:2	1	Y	Christiansen 2018
K2-138	d	e	3:2	1	Y	Christiansen 2018
K2-138	e	f	3:2	1	Y	Christiansen 2018
Kepler-223	b	c	4:3	1	N	Mills 2016
Kepler-223	c	d	3:2	1	Y	Mills 2016
Kepler-223	d	e	4:3	1	N	Mills 2016
Kepler-80	d	e	3:2	1	Y	MacDonald 2016
Kepler-80	e	b	3:2	1	Y	MacDonald 2016
Kepler-80	b	c	4:3	1	N	MacDonald 2016
Kepler-80	c	f	3:2	1	Y	MacDonald 2016
TOI-178	c	d	2:1	1	Y	Leleu 2021
TOI-178	d	e	3:2	1	Y	Leleu 2021
TOI-178	e	f	3:2	1	Y	Leleu 2021
TOI-178	f	g	4:3	1	N	Leleu 2021
GJ-876	c	b	2:1	1	Y	Rivera 2010
GJ-876	b	e	2:1	1	Y	Rivera 2010
HD 158259	b	c	3:2	1	Y	Hara 2020
HD 158259	c	d	3:2	1	Y	Hara 2020
HD 158259	d	e	3:2	1	Y	Hara 2020